LIME Cost/Quality Scaling on High-Dimensional Tabular Data Across Multiple Models

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Abstract—Local Interpretable Model-agnostic Explanations (LIME) is widely used to explain black-box classifiers, but there is little guidance on how its cost and explanation quality scale together as dimensionality grows. We ran a full factorial study on the UCI Human Activity Recognition dataset, crossing three models (logistic regression, random forest and XGBoost), four perturbed-feature counts from 50 to 500 and five sample sizes from 500 to 10,000, over 30 explained cases. Runtime grew almost linearly with sample size, and the marginal cost per 1,000 samples rose from 9–17 ms at 50 features to 78–86 ms at 500. Held-out surrogate fidelity levelled off by 5,000 samples and stayed low (R^2 of about 0.3 or less), whereas agreement between repeated explanations kept improving up to 10,000 samples at a steeply rising marginal cost. At 300 features, model choice mattered more for quality than for cost: runtimes differed by 17%, while XGBoost agreed significantly more closely with a 50,000-sample reference (0.86) than random forest (0.69) or logistic regression (0.61).

Keywords—explainable AI, LIME, local surrogate models, explanation stability, computational cost, high-dimensional tabular data

I. Introduction

Local Interpretable Model-agnostic Explanations (LIME) [1] explains an individual prediction by perturbing the instance, querying the black-box model on the perturbed samples and fitting a weighted linear surrogate whose largest coefficients form the explanation. Its runtime therefore depends on the number of samples N and on the model's inference cost. Its explanation quality is expected to suffer as the number of perturbed features d grows, because a fixed budget covers a larger space more thinly.

Prior work has examined LIME's instability and proposed variants that reduce it, but offers little guidance on how cost and quality scale together with dimensionality and model complexity (Section II-B). This paper measures LIME's runtime against the stability and fidelity of its explanations for three models on a high-dimensional tabular dataset, to find where extra computation stops improving quality. It asks: (RQ1) how does runtime scale with N and d? (RQ2) how does explanation quality change with N and d? and (RQ3) how does the choice of black-box model change the cost/quality trade-off?

II. Background and Related Work

A. LIME in Brief

For tabular data, LIME [1], [2] first discretises each continuous feature into quartiles computed from a background sample of training data. It then draws N perturbed samples from this background distribution, represents each sample by whether each feature falls in the same quartile as the explained instance, and weights each sample with an exponential kernel on its distance from the instance. A ridge regression surrogate is fitted to the model's predictions on the weighted samples, and the k features with the largest coefficients form the explanation. As sampling is random, two runs on the same instance can return different explanations (stability); because the surrogate is linear, it may fit the model's local behaviour poorly (fidelity).

B. Related Work

Where LIME's runtime has been reported, it has been a side result rather than a systematic measurement, and not on high-dimensional tabular data [3]–[5]. Second, studies show that more samples improve stability at a higher computational cost, but this trade-off has not been shown across models on high-dimensional tabular data [6]–[8]. Third, tabular LIME has been evaluated for the usefulness of its explanations, but its cost has not been measured alongside that evaluation [9]. This study addresses all three gaps by timing LIME systematically on a high-dimensional tabular dataset, across several models, jointly with stability and fidelity.

C. Choice of Models

Each of the three models represents a widely used tabular classifier family with a different inference cost and decision surface. Logistic regression [10] is linear in its log-odds and cheapest to query, giving a simple baseline. Random forest [11] averages many decision trees trained on bootstrap samples (bagging), giving a piecewise-constant ensemble output. This model provides adequate contrast as a more complex model. XGBoost [12] builds trees sequentially by gradient boosting and is a common default for tabular prediction, with sharper, more interaction-rich decision boundaries. This is a non-linear complex model, thus benefitting our comparison with more diversity. Comparing them separates the model's effect on LIME's cost from its effect on explanation quality

D. Choice of Dataset

The UCI Human Activity Recognition (HAR) dataset [13] contains 561 continuous features derived from a waist-mounted smartphone's accelerometer and gyroscope, recorded while 30 participants performed six daily living activities. This is high-dimensional compared with typical tabular datasets, and its features are dense and continuous, which suits LIME's quartile discretisation. The official split assigns each participant to either the training or the test set [14], so leakage is avoided; the original

authors report about 96% accuracy on this split [13]. Covertype, considered initially, was rejected because its 54 features are too few to study dimensionality [15].

III. Methodology

A. Study Design

A full factorial experiment crosses three models, four perturbed-feature counts d ∈ {50, 100, 300, 500} and five sample sizes N ∈ {500, 1,000, 2,500, 5,000, 10,000}, with five repeats per cell, over 30 explained cases. The case is the unit of analysis: repeats are averaged within a case first. Each case also receives two reference explanations at N = 50,000 for each d.

B. Dataset and Explained Cases

The official subject-based split is used: 21 participants (7,352 rows) for training and 9 participants (2,947 rows) for testing. Thirty test cases, five per class, were selected from the true labels only, without reference to any model's predictions, so that no model is favoured by rows that happen to suit it. At most four cases come from any one participant, so that no single person's near-identical rows dominate. Every model explains each case's true class, so all three models answer the same question about the same row.

C. Models

Table I lists the model settings. Each model is trained once on all 561 features and is not retrained for each value of *d*. Hyperparameters are neither tuned nor matched across models, since the aim is to compare LIME's behaviour on representative model families rather than to maximise accuracy. The random state is 42, and all inference is single-threaded.

TABLE I. Model Settings

Model	Library	Settings	Test acc.
LR	scikit-learn 1.7.2	StandardScaler + LR pipeline; L2, C = 1, lbfgs, max_iter = 1000	95.5%
RF	scikit-learn 1.7.2	100 trees	92.8%
XGB	XGBoost 3.2.0	250 trees, depth 6, learning rate 0.1, subsample 0.9	94.1%

D. Perturbed Features and Sample Sizes

The perturbed features are nested prefixes (50 ⊂ 100 ⊂ 300 ⊂ 500) of one random permutation of the 561 features, seeded with 42, so a larger *d* only adds features to a smaller one. LIME perturbs only these *d* features; the rest are frozen at the explained row's values, so the model always receives all 561 inputs and stays fixed while only LIME's setting varies. The sample sizes bracket LIME's default of 5,000, giving *N*:*d* ratios from 1:1 to 200:1. The single cell with $N \le d$ ($N = d = 500$) is flagged and excluded from trend lines and primary tests.

E. LIME Configuration

Explanations use LimeTabularExplainer [2] from LIME 0.2.0.1 [16], with quartile discretisation, Euclidean distance and a ridge surrogate with α = 1. The exponential kernel has the library-default width of $0.75\sqrt{d}$, because the explainer sees only the d perturbed columns. The background data are 2,000 random training rows, about 500 per quartile, which is enough for stable bin edges at low cost. Each explanation reports k = 10 features, so feature_selection = "auto" resolves to highest_weights. Per-case seeds from NumPy's SeedSequence make every explanation reproducible; wall-clock times, by nature, are not.

F. Evaluation Metrics

Cost is the wall-clock time of each explain_instance() call, measured with perf_counter. Stages were not timed separately: sampling and kernel weighting scale with N·d, and only prediction depends on the model.

Held-out R² (fidelity) scores the surrogate against the model on 5,000 held-out perturbations per case and d, with kernel weighting. The held-out set is shared across all N, repeats and models, so comparisons are paired. Kernel-weighted RMSE is also recorded, but it is not comparable across models. Random forest's output barely moves locally (weighted SD ≈ 0.06, against 0.20–0.36 for the other models at d ≥ 300), so its absolute errors are small regardless of fit. R^2 is excluded where the output is near-constant (weighted SD < 0.01). This affects 21 of 120 case × d combinations for LR and 33 for XGB, all at d ≤ 100, so at d ≥ 300 every model's R^2 covers all 30 cases.

Repeat Jaccard (stability) is the mean top-5 Jaccard overlap across the ten pairs of the five repeats. *Reference Jaccard* (convergence) is the top-5 overlap with the two N = 50,000 references. Both metrics use the top 5 of the 10 reported features to emphasise agreement among the most influential features. References for a model × d combination are accepted only if they agree with each other at a mean Jaccard of at least 4/6, the value when two top-5 sets share four features. All 12 combinations passed (0.80–0.97).

G. Measurement Protocol and Analysis

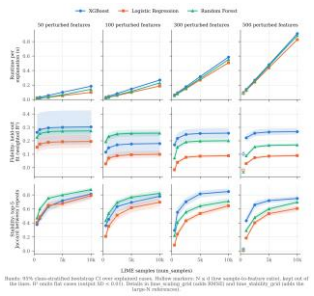
Models run interleaved in randomised blocks, one block per (case, *d*, *N*, repeat), with model order shuffled within each block, so that machine drift cannot favour one model. All runs share one process on one machine. BLAS and OpenMP are single-threaded; warm-up runs are untimed, and explainer construction and scoring fall outside the timed region. Experiments ran on Python 3.13.7 on an Apple Silicon Mac (8 cores) with global seed 42.

Uncertainty is estimated with a class-stratified case bootstrap (2,000 resamples, 95% percentile intervals), with participant-clustered intervals as a conservative check. Differences are tested with paired sign-flip permutation tests (10,000 flips) and Holm correction [17]. The pre-specified primary family has 36 tests: the three model pairs at each d at N = 5,000, for runtime, RMSE and reference Jaccard. RMSE stays in that family as pre-specified but is not interpreted. R^2 comparisons are Holm-corrected across all model-pair tests, and comparisons between values of d within a model are corrected within a separate family of 54 tests. Runtime scaling per model is summarised by the fit $\log t = a + b \log N + c \log d$.

IV. Results

Fig. 1 shows runtime and the quality measures against N for each model and d; Table II gives the main values at LIME's default of N = 5,000.

Fig. 1.



Runtime, held-out fidelity and top-5 agreement against sample size N for each model and perturbed-feature count d.

A. Cost (RQ1)

Runtime grew almost linearly with N on top of a fixed per-explanation overhead (straight-line fits, $R^2 \geq 0.993$ for every model × d). In the log-log fit, b ≈ 0.72–0.77, so each doubling of N multiplied runtime by 1.65–1.70, below 2 because of the overhead; c ≈ 0.66 for XGBoost, 0.83 for random forest and 0.90 for logistic regression. The marginal cost per 1,000 samples, the slope of the linear fit, grew from 9–17 ms at d = 50 to 78–86 ms at d = 500. Logistic regression was the fastest model and XGBoost the slowest in every setting, but the gap narrowed from 1.85× at d = 50 to 1.10× at d = 500 (N = 10,000).

B. Explanation Quality (RQ2)

Held-out R^2 levelled off by N = 5,000 (changing by under 0.005 from 5,000 to 10,000) and stayed low (0.08–0.30). Repeat Jaccard, by contrast, kept improving up to N = 10,000 at a rising cost: for XGBoost at d = 500, each 0.01 gain cost 0.006 s between N = 1,000 and 2,500 but 0.137 s between 5,000 and 10,000. At N = d = 500, repeat Jaccard was only 0.004–0.029. It declined significantly with d for RF (d = 50 and 100 each differed from 300 and 500) and for LR (50 vs 300), but not detectably, and not monotonically, for XGBoost.

C. Model Effects (RQ3)

At d = 300, XGBoost had the highest reference Jaccard, significantly above both other models after Holm correction and under participant-clustered intervals, while runtimes differed by only 17%. At d = 500 the ordering held, but no difference survived correction ($p_{Holm} \geq 0.37$). At d ≤ 100, random forest was highest, significantly above logistic regression only at d = 100 ($p_{Holm} = 0.009$), while runtimes differed by up to 1.9×. At low d, therefore, the model affected cost more than quality.

Logistic regression, despite the highest test accuracy, had the lowest reference Jaccard and held-out R^2 at every d. Its R^2 was significantly below both other models at d ≥ 300 ($p_{Holm} = 0.036$).

TABLE II. KEY RESULTS AT N = 5,000

Measure	d	LR	RF	XGB
Runtime (s)	500	0.438	0.465	0.480
Held-out R^2	300	0.085	0.198	0.254
Held-out R^2	500	0.085	0.165	0.265
Repeat Jaccard	50	0.677	0.798	0.719
Repeat Jaccard	300	0.534	0.642	0.814
Reference Jaccard	50	0.729	0.834	0.779
Reference Jaccard	100	0.665	0.825	0.737
Reference Jaccard	300	0.609	0.692	0.855
Reference Jaccard	500	0.610	0.685	0.761

V. DISCUSSION

A. Why the Runtime Gap Narrows at High d

LIME's sampling and kernel weighting scale with $N \cdot d$, whereas inference does not depend on d because the model always receives all 561 inputs. At high d, LIME's overhead therefore plausibly outweighs inference, shrinking the gap between models from 1.85× to 1.10×. XGBoost's lower d-exponent ($c \approx 0.66$) is consistent with this: its larger, d-independent inference cost forms a bigger share of the total, so the d-dependent part matters relatively less. Since individual stages were not timed, this explanation remains to be confirmed.

B. Fidelity Plateau versus Continued Stability Gains

Fidelity is capped by how well a linear surrogate on binary quartile indicators can approximate the model locally. Once its coefficients are well estimated, more samples cannot remove this structural error, so R^2 levels off. Stability depends on the variance of those coefficients, which keeps shrinking as N grows. This agrees with the link between sample size and stability described in [6], [8], and adds that the gain is not matched by fidelity and becomes steeply more expensive. At $N = d = 500$ the surrogate is barely determined.

C. The Most Accurate Model Gave the Weakest Explanations

Accuracy describes behaviour on real data, whereas LIME probes the model on synthetic perturbations, so the two need not agree. Saturation does not explain LR's weakness: XGBoost's output is saturated on more cases (24 vs 20 of 30), yet XGBoost explains best at $d \geq 300$. One plausible, untested reason is that LR depends on many correlated features at once, so small sampling differences reorder its top features and a quartile-based surrogate fits it poorly.

VI. LIMITATIONS

The default kernel width of $0.75\sqrt{d}$ changes with d, so part of the apparent effect of d may come from the kernel. The perturbed features are random rather than the most important ones, so at low d LIME may perturb features the model barely uses, which may explain the near-constant outputs. Fixed-kernel and ranked-subset variants exist in the code but were not run in full. Only 30 cases from 9 test participants on one dataset were explained. LIME samples each feature independently, so its perturbations combine sensor readings that cannot occur together and query the model away from its training data. Each model used one untuned configuration, so the findings describe these models rather than their families in general. Absolute runtimes come from one single-threaded machine and will not transfer to other hardware, though the scaling exponents and relative comparisons should.

VII. CONCLUSION

Beyond LIME's default budget, extra samples buy repeatability rather than faithfulness, and at $d \geq 300$ the model matters far more for explanation quality than for cost. For $d \geq 300$, use at least $N = 5{,}000$ and avoid $N \leq d$. Users who need faithful explanations should measure fidelity directly rather than raise the sample budget. Future work should run the fixed-kernel and ranked-subset variants on other datasets and models.

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